

HealthConnect Clinic Experience Lab: Analytics Section

Executive Summary of Analytical Findings

The analytics track for the HealthConnect Clinic Experience Lab evaluated 5,000 rigorous operational and transactional records to establish baseline clinical performance metrics, patient behavioral patterns, and systemic friction points.

Core Operational Benchmarks

Metric Name	Observed Value	Operational Implication
No-Show Rate	48.46%	Nearly half of scheduled appointments result in unfulfilled slot capacity, representing severe resource waste.
Attendance Rate	46.28%	Actual patient fulfillment trails structural booking volume due to high attrition.
Mean Booking Lead Time	29.64 Days	Extended windows between booking and appointment dates exponentially increase drop-off likelihood.
Average Wait Time	24.19 Minutes	On-site processing delays contribute to patient friction and dissatisfaction.
Average Travel Distance	10.11 Kilometers	Geographic dispersion acts as a physical barrier to consistent clinic attendance.

Methodological Rigor & Data Quality Audit

- **Integrity Verification:** Confirmed 100% Primary Key uniqueness across all records.
- **Error Mitigation:** Executed missing value imputations and eliminated duplicate entry artifacts to ensure pristine dataset reliability.
- **Data Hygiene:** Validated data distributions against known healthcare administrative norms to safeguard executive reporting integrity.

Predictive Modeling & Feature Importance

Analytical modeling identified primary behavioral and logistical triggers governing patient attendance:

1. **Booking Horizon Decay:** Lead times exceeding 14 days correlate with a 3x increase in default probability.
2. **Historical Attrition Weighting:** Prior no-show counts serve as the single strongest statistical predictor of future default behavior.
3. Distance decay curves confirm that patients residing beyond the 10.11 km threshold experience sharply elevated drop-off rates without proactive digital touchpoints.